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Review text:

The degree/diameter problem for directed graphs asks to find the maximum order of a digraph with maximum out-degree d and diameter k , for which a natural upper bound is the Moore bound $1 + d + d^2 + \cdots + d^k$. Digraphs attaining this bound are called Moore digraphs, and they do not exist unless $d = 1$ or $k = 1$. So, a natural problem is to investigate almost Moore digraphs, also known as (d, k) -digraphs, which are diregular digraphs of degree $d > 1$ and diameter $k > 1$ whose order is one less than the Moore bound; see Miller–Širáň [Electron. J. Combin. **DS14** (2005), 61 pp.; MR4336216] for a dynamic survey of the area. Almost Moore digraphs exist for $k = 2$; see Fiol–Alegre–Yebra [ACM SIGARCH Comput. Archit. News **11** (1983), 174–177] and Gimbert [Discrete Math. **231** (2001), no. 1–3, 177–190; MR1821958]. It is believed that they do not exist for $k \geq 3$; this has been verified for $d = 2$ by Miller–Friš [In: R. S. Rees (Ed.), Graphs, Matrices, and Designs, Lecture Notes in Pure and Appl. Math., vol. 139, Marcel Dekker, New York, 1993, 269–278; MR1209196], for $d = 3$ by Baskoro–Miller–Širáň–Sutton [J. Graph Theory **48** (2005), no. 2, 112–126; MR2110581], for $k = 3$ by Conde–Gimbert–González–Miret–Moreno [Electron. J. Combin. **15** (2008), no. 1, #R87, 14 pp.; MR2426150] and for $k = 4$ by Conde–Gimbert–González–Miret–Moreno [Electron. J. Combin. **20** (2013), no. 1, #P75, 14 pp.; MR3040637].

It is known that in any (d, k) -digraph G , each vertex $v \in V(G)$ has a unique repeat $r(v) \in V(G)$, characterized by the property that there are exactly two $v \rightarrow r(v)$ walks in G of length at most k , and that the map $r: V(G) \rightarrow V(G)$ is

an automorphism of G , as shown by Baskoro–Miller–Plesník [Graphs Combin. **14** (1998), no. 2, 109–119; MR1628113]. A vertex v is a self-repeat if $r(v) = v$. López–Messegue–Miret [Electron. J. Combin. **30** (2023), no. 1, #P1.56, 15 pp.; MR4564832] showed that $(4, k)$ - and $(5, k)$ -digraphs with self-repeats do not exist for $k \geq 5$.

In this article, the authors show that, for $d \geq 6$, there are no (d, k) -digraphs with self-repeats when $k \geq 2(d - 1)$ and k is odd, or when $k \geq 2(d - 1)^2$ and k is even. The key structural ingredient, which extends a technique of Sillasen [Electron. J. Graph Theory Appl. **3** (2015), no. 1, 1–7; MR3333722] of studying the subdigraphs induced by the vertices fixed by powers of r , is that any (d, k) -digraph with self-repeats contains a subdigraph that is itself an almost Moore digraph of some degree $d' \leq d$ and diameter k in which every vertex has “order” at most $d' - 1$, where the order of a vertex v is the least integer $t > 0$ such that $r^t(v) = v$.

Let $F_{i,k}(x) = \Phi_i(1 + x + \cdots + x^k)$ for integers $i > 2$ and $k > 1$, where Φ_i is the minimal polynomial over $\mathbb{Q}$ of a primitive i th root of unity—these polynomials arise in the factorization of the characteristic polynomial of the adjacency matrix of a (d, k) -digraph. The cyclotomic conjecture of Gimbert [Discrete Math. **197/198** (1999), 375–391; MR1674874] states that $F_{i,k}(x)$ is irreducible over $\mathbb{Q}$, unless k is even and i divides $k + 2$, or k is odd, i is even and i divides $2(k + 2)$, in which cases $F_{i,k}(x)$ has exactly two irreducible factors. It is known from the work of Conde–Gimbert–González–Miller–Miret [European J. Combin. **39** (2014), 170–177; MR3168522] that if the conjecture holds for every index i with $2 < i \leq d + d^2 + \cdots + d^k$, then (d, k) -digraphs do not exist. Moreover, for a (d', k) -digraph in which every vertex has order at most $d' - 1$ under r , only the indices $i \leq d' - 1$ are relevant. In this paper, the authors verify the conjecture computationally for $2 < i \leq 12$ and $4 < k \leq 200$; together with their main theorem and a direct search for suitable primes for the even diameters $202 \leq k \leq 240$ when $d = 12$, this yields the nonexistence of (d, k) -digraphs with self-repeats for $6 \leq d \leq 12$ and $k > 2$. After the publication of this work, a proof of the cyclotomic conjecture in full generality, and hence of the nonexistence of (d, k) -digraphs for all $d > 1$ and $k > 2$, has been announced by Kaur–Kumar [“A proof of the cyclotomic conjecture and the non-existence of almost Moore digraphs”, preprint (2026), arXiv:2608.08997].

Some minor typos: in the statement of Conjecture 1(ii), the hypothesis that i is even is missing; in the caption of Figure 1, “the case $n_j(1)$ for all j ” should read “the case $n(j) = 1$ for all j ”; in the caption of Figure 2, “except for on subindex” should read “except for one subindex”; and in reference [19], the paper number should be #P1.56 instead of #P156.